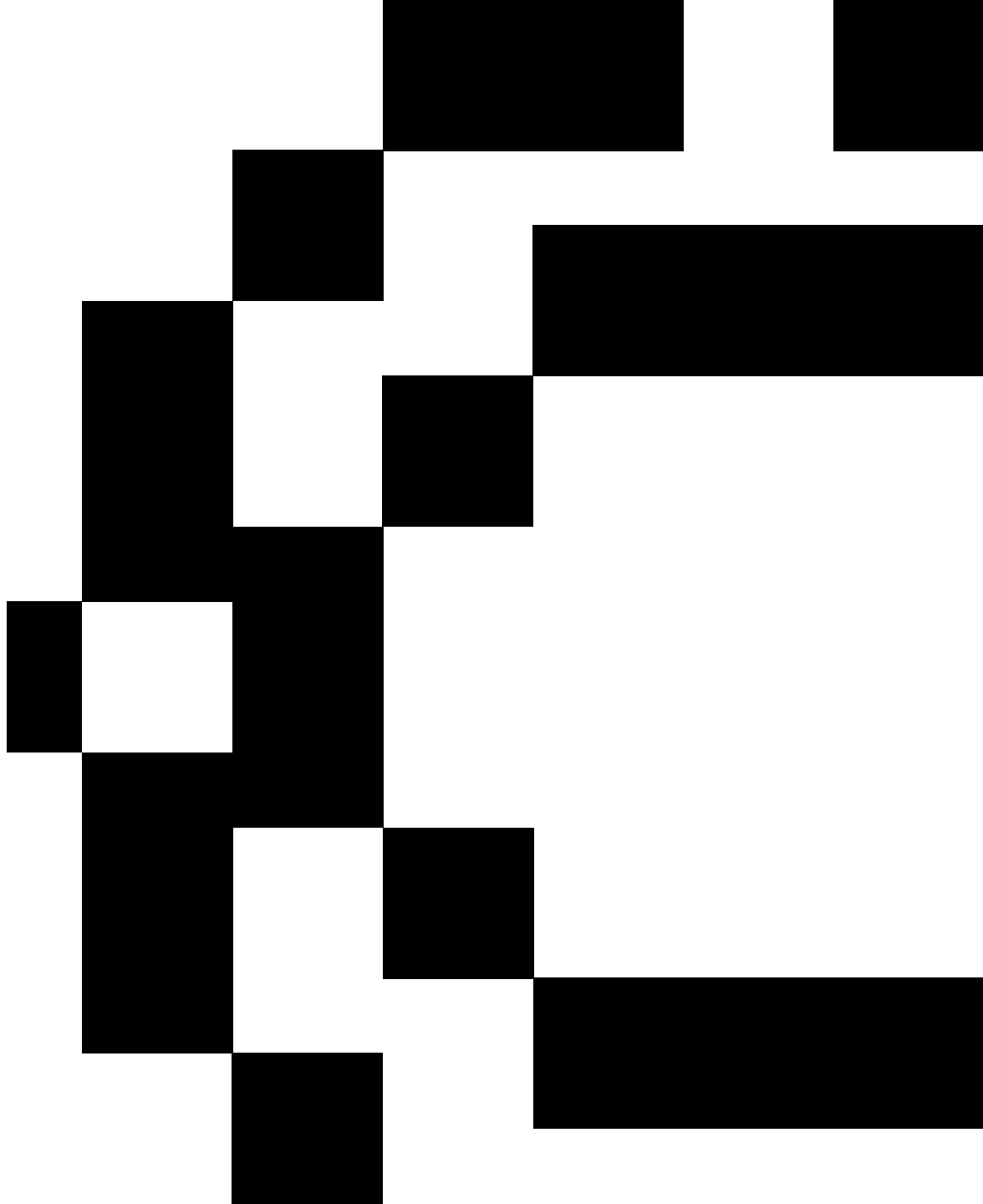
NOVA IMS INFORMATION MANAGEMENT SCHOOL

# Computation I / Introduction to Programming

## Practicals Week 05 - Conditionals



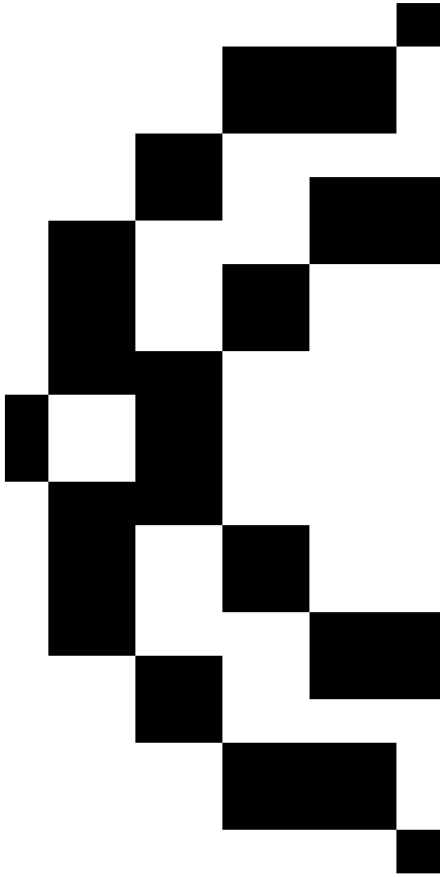
# Exercise 1

Ask the user to type their favourite colour. If the colour is yellow, tell them they have terrible taste and you do not understand them. Otherwise, print a sentence saying you can see why that colour would be appealing to them.

**Examples**

- `Input:  color = "yellow"`
- `Output: You have terrible taste and I do not understand you.`

- `Input:  color = "blue"`
- `Output: I can see why blue would be appealing to you`



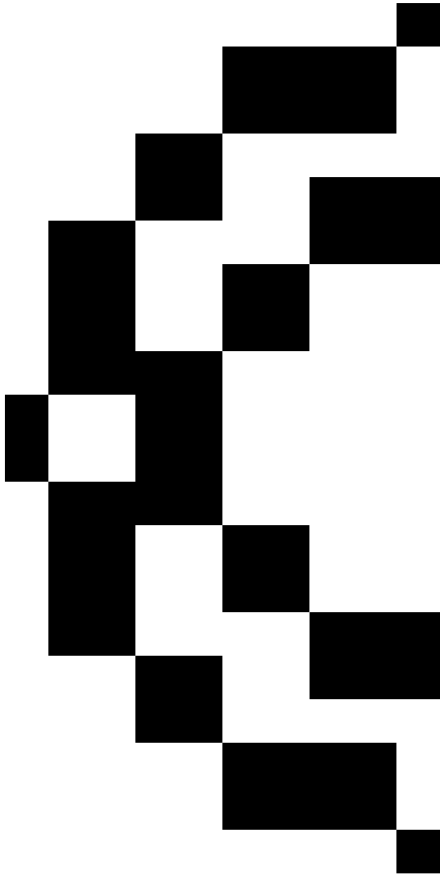
# Exercise 2

Write a program that reads an integer from the user. If the number is zero or positive, calculate and print its square root. If the number is negative, print a message saying the number inserted is negative.

**Examples**

- `Input:  number = 16`
- `Output: The square root of 16 is 4.0`

- `Input:  number = -9`
- `Output: The number is negative`



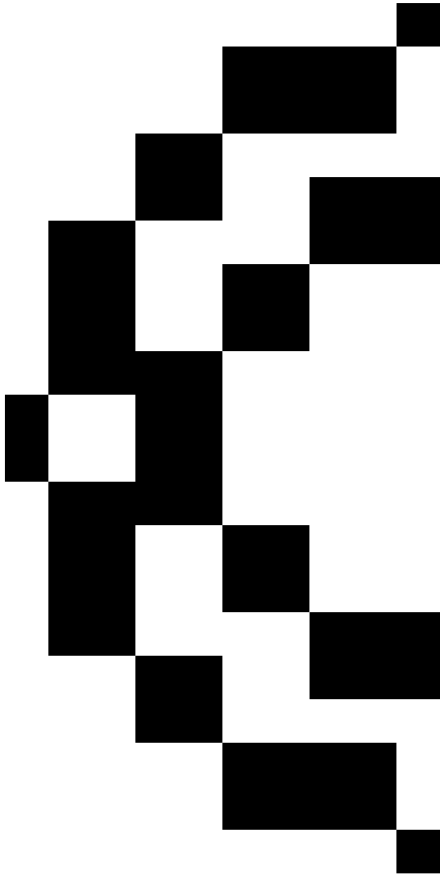
# Exercise 3

Ask the user for their name, then ask them (as a Boolean, typed as `True` or `False`) whether milk should be added before the cereal. If they answer `True`, inform them they will not pass Computation I. If they answer `False`, tell them they have some chance of passing. If they type anything else, tell them they did not enter a Boolean value.

**Examples**

- `Input:  name = "Alex", answer = "True"`
- `Output: Dear Alex, you cannot pass Computation I, good luck with your life.`

- `Input:  name = "Sam", answer = "False"`
- `Output: Dear Sam, congrats, you have chance to pass this class!`



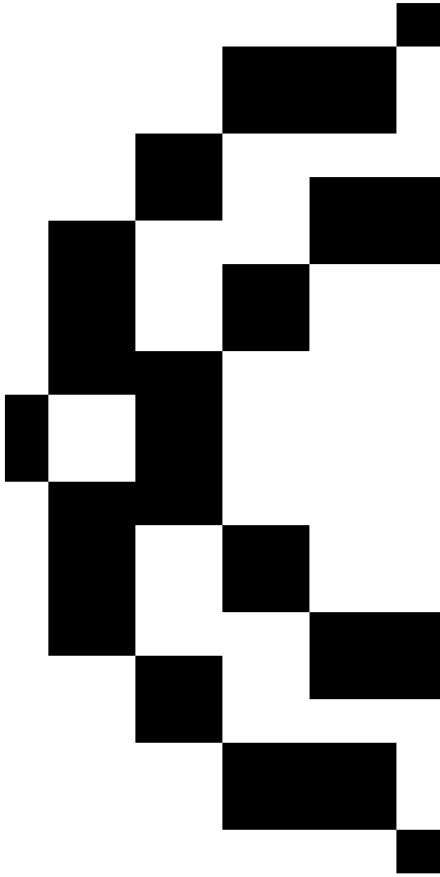
# Exercise 4

Ask the user for two numbers, `a` and `b`. Then ask them to choose an operation:
`1` for multiplication, `2` for addition, `3` for `a` raised to the power of `b`.
Print the result of the chosen operation.

**Examples**

- `Input:  a = 4, b = 2, choice = 1`
- `Output: The result is: 8.0`

- `Input:  a = 4, b = 2, choice = 3`
- `Output: The result is: 16.0`

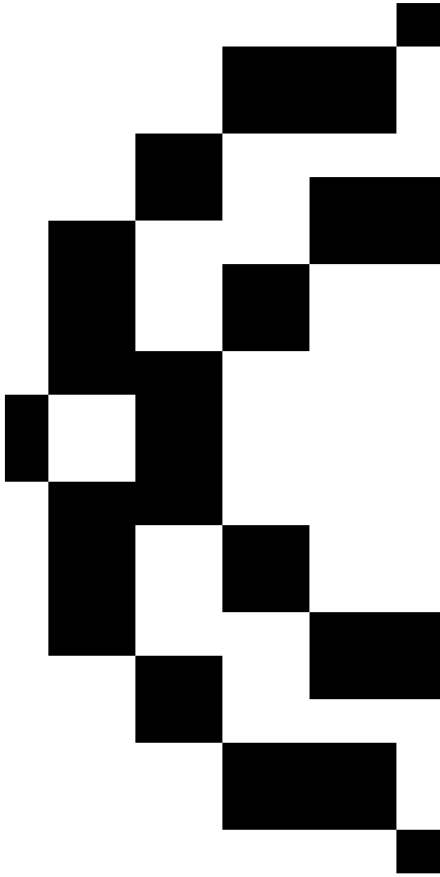


# Exercise 5

Gertrude went shopping for soda again. Ask the user how many sodas she bought and how much she paid for each one. Calculate and print the total. If she paid more than €10 in total, tell her she is spending too much; otherwise, tell her she is within budget. Separately, if she bought more than 5 sodas, warn her that she is drinking too many — regardless of what was printed about the budget.

**Example**

```
Input:  number = 6, price = 2.00
Output: Gertrude spent 12.0 on sodas
        She is spending too much!
        Gertrude is drinking an unhealthy amount of soda...
```



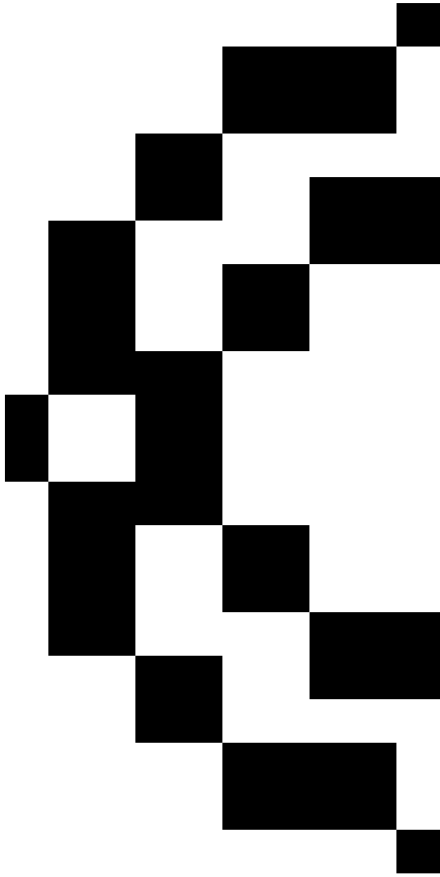
# Exercise 6

Ask the user to type an integer. If the number is smaller than 1000, print that the number is small. Otherwise, print how many digits the number contains.

**Examples**

- `Input:  integer = 42`
- `Output: Your number is small`

- `Input:  integer = 48213`
- `Output: 48213 has 5 digits`



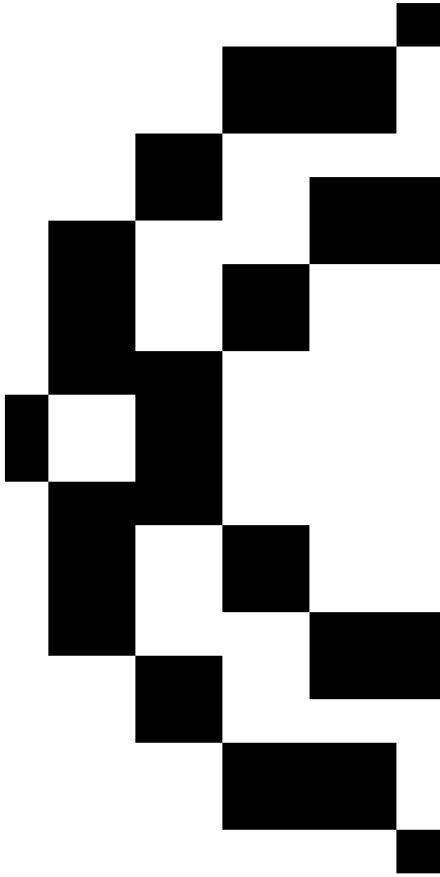
# Exercise 7

Ask the user for their phone number, including the country code at the start.
Using the table below, print the name of the country if the code is recognised,
or a message saying the country is not recognised otherwise.
+351 → Portugal
+90 → Turkey
+354 → Iceland

**Examples**

- `Input:  phone = "+351912345678"`
- `Output: Your number is Portuguese!`

- `Input:  phone = "+90212345678"`
- `Output: Your number is Turkish!`



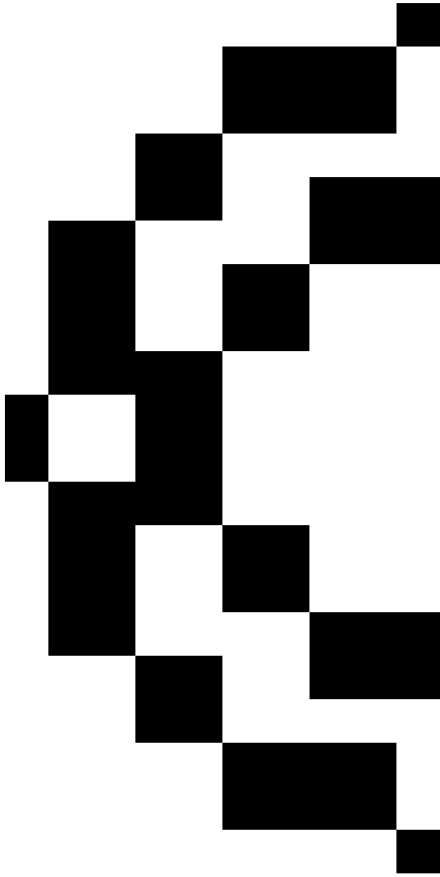
# Exercise 8

Ask the user for their name. If it ends in `"a"`, replace the final `"a"` with `"inha"`. If it ends in `"o"`, replace the final `"o"` with `"inho"`. Otherwise, leave the name unchanged. Print a friendly greeting using the (possibly changed) name.

**Examples**

- `Input:  name = "Carolina"`
- `Output: Nice to meet you Carolininha`

- `Input:  name = "Bruno"`
- `Output: Nice to meet you Bruninho`



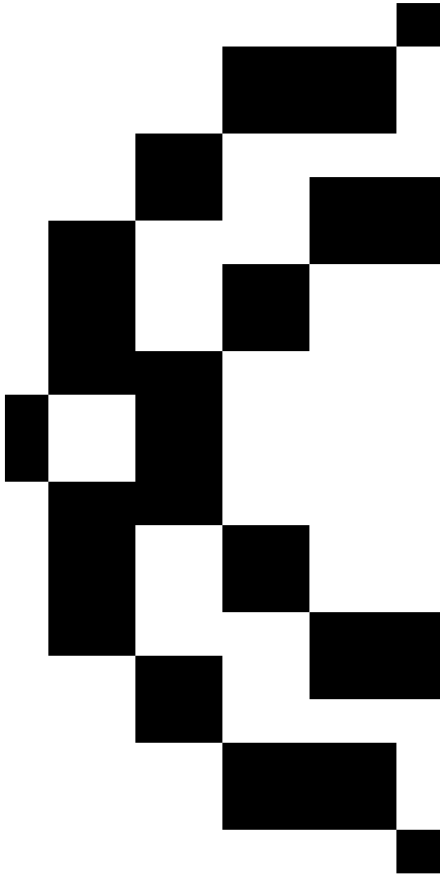
# Exercise 9

Ask the user to say something about Nova. If their sentence contains the word "Nova" (in any capitalisation), replace it with "Universidade Nova de Lisboa" and print the updated sentence. Otherwise, ask them if they are sure they are talking about Nova.

**Examples**

- `Input:  sentence = "I study at Nova and I love it"`
- `Output: i study at universidade nova de lisboa and i love it`

- `Input:  sentence = "I like pizza"`
- `Output: Are you sure you are talking about Nova?`



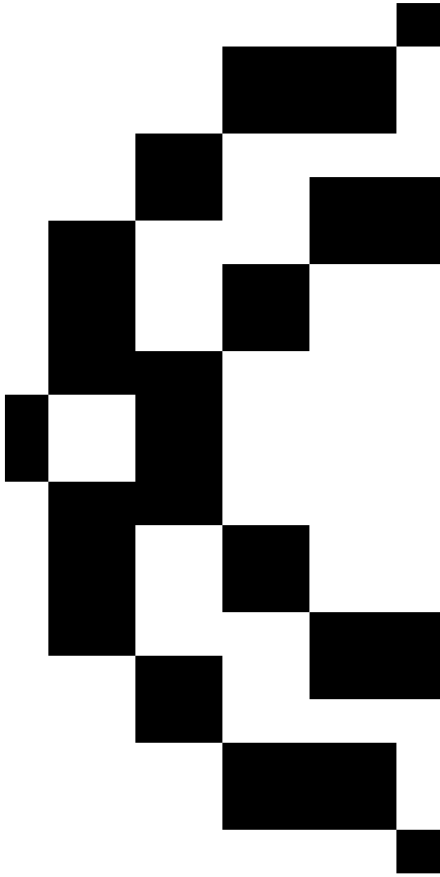
# Exercise 10

Ask the user for the name of their pet, then ask them to choose its species:
1 for dog, 2 for cat, 3 for any other animal. Print the cost of a vet
visit: €50 for a dog, €40 for a cat, and €60 for anything else.

**Examples**

- Input: `pet = "Rex", species = 1`
- Output: `The visit will cost you 50€`

- Input: `pet = "Whiskers", species = 2`
- Output: `The visit will cost you 40€`



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# Thank you

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